# Final Report

## Monitoring & Assessment

Standardized framework for gamified telemetry integration, cognitive metrics, clinical context, and traceability.

Neuropsychological assessment document generated with Teko digital biomarkers.

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Standardized final report structure with telemetry synthesis, behavioral analysis, and clinical conduct.

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# Identification

**Patient's Name**
Lucas Gabriel Silva

**Age**
8 years and 4 months

**Gender**
Male

**Guardian**
Helena Silva (Mother)

**Assessment Date**
10/01/2026

**Responsible Psychologist**
Dr. Mariana Costa

**Professional License (CRP)**
CRP 06/142589

**Clinic / Institution**
Teko Neuropsychology Center

**Battery Completion Status**
Complete Battery (100% completed)

**Completed Games**

- Goalkeeper (Visuomotor Reaction Time)
- Tap Fast (Go/No-Go Inhibitory Control)
- Forest Photographer (Sustained Attention and Fatigue)

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# Reason for Assessment

### Assessment Objective

Map digital biomarkers of attentional regulation, visuomotor reaction time, and inhibitory control through Teko's standardized serious games battery.

### Main Complaint

School and family reports highlighting motor restlessness, inattention in long-duration tasks, and frequent impulsivity during homework completion.

### Initial Context Summary

Child without prior consolidated diagnoses. Fluctuating academic performance in 3rd grade with good sociability and intellectual curiosity.

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# Methodology

**Neuropsychological Assessment Based on Digital Biomarkers**
The Teko methodology combines structured computerized tasks with continuous registration of visuomotor interaction. By replacing paper tests with adaptive games, the child's natural behavior is captured in real time with millisecond precision.

## Methodological Description

The Teko battery utilizes serious games with continuous telemetry of micro-movements, visuomotor response latency, and precision rates. The platform evaluates cognitive profiles in a playful environment, recording millisecond-accurate metrics without burdening the child.

## Session Conditions & Technical Quality

In-person assessment in a structured and quiet office setting during morning hours. The child showed high cooperation, strong motivation, and full adherence to game instructions.

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# Model Variables

### 1. Response Variability

Measures millisecond-level regularity of motor responses to rapid visual stimuli in the Goalkeeper game.

### 2. Inhibitory Control

Evaluates motor braking and deceleration capability during high-speed No-Go stimuli in the Tap Fast game.

### 3. Sustained Attention

Assesses focus stability and perseverance over time through accuracy and lapses in Forest Photographer.

### 4. Age Standardization

Statistical standardization and normalization of digital biomarkers according to normative peer age groups.

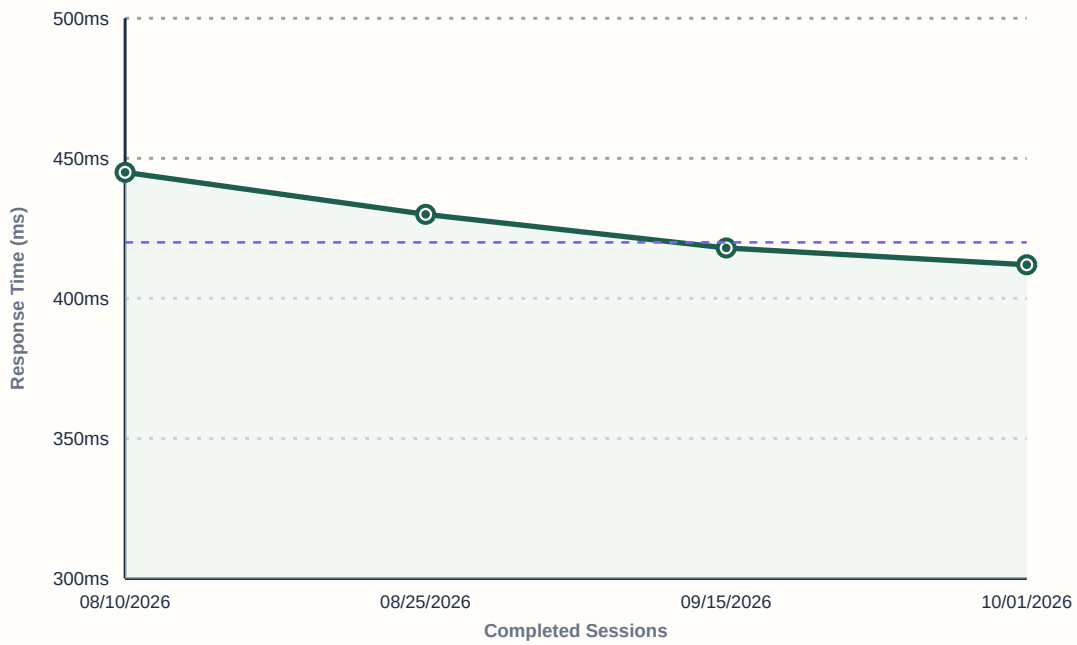
### Variables Effectively Used in Assessment

All biometric variables for reaction time, inhibitory control, and sustained attention showed full sampling consistency and telemetry integrity.

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# Game Results

### Goalkeeper Game: Visuomotor Response Time across Sessions




### Clinical Synthesis & Evolution: Goalkeeper Game

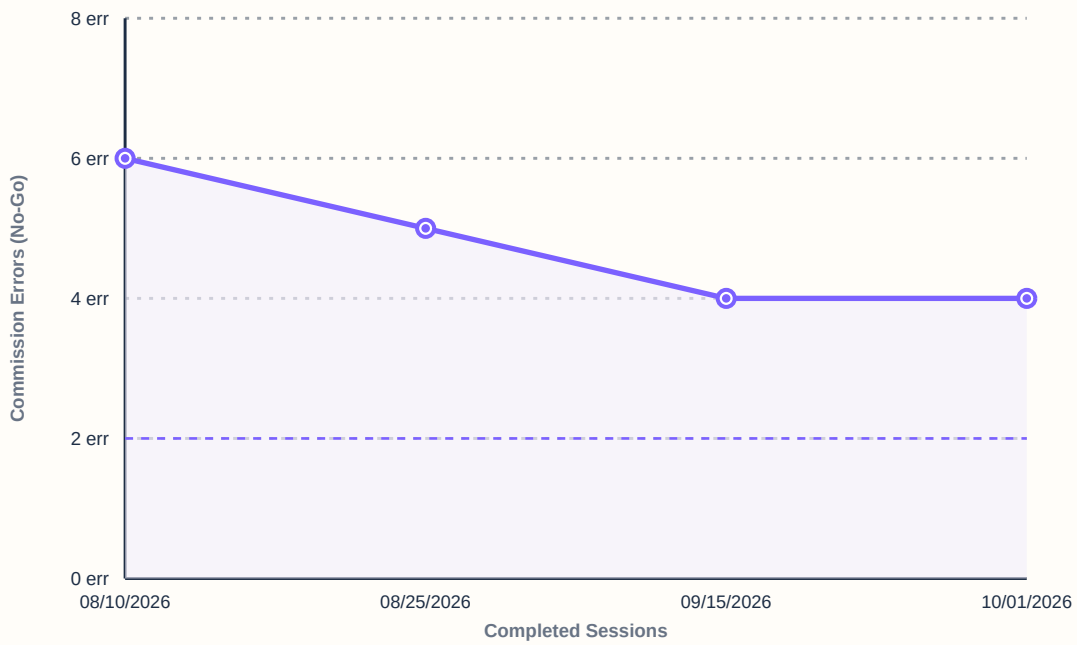
**Neurofunctional Analysis & Performance Opinion:**
Moderate fluctuation in response latencies, with quick reactions interspersed with transient lapses in the final blocks. Across sessions, latency stabilized from 445ms to 412ms, reflecting positive adaptation and reduced attentional jitter.

**Visuomotor Trajectory Evolution:** Continuous telemetry demonstrates that the child establishes early readiness and stability, with performance variance concentrated during blocks demanding heightened temporal precision. The progressive latency reduction observed across sessions reflects positive adaptive capacity under targeted intervention.

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# Game Results

**Tap Fast Game: No-Go Inhibitory Commission Errors across Sessions**




**Clinical Synthesis & Evolution: Tap Fast Game**

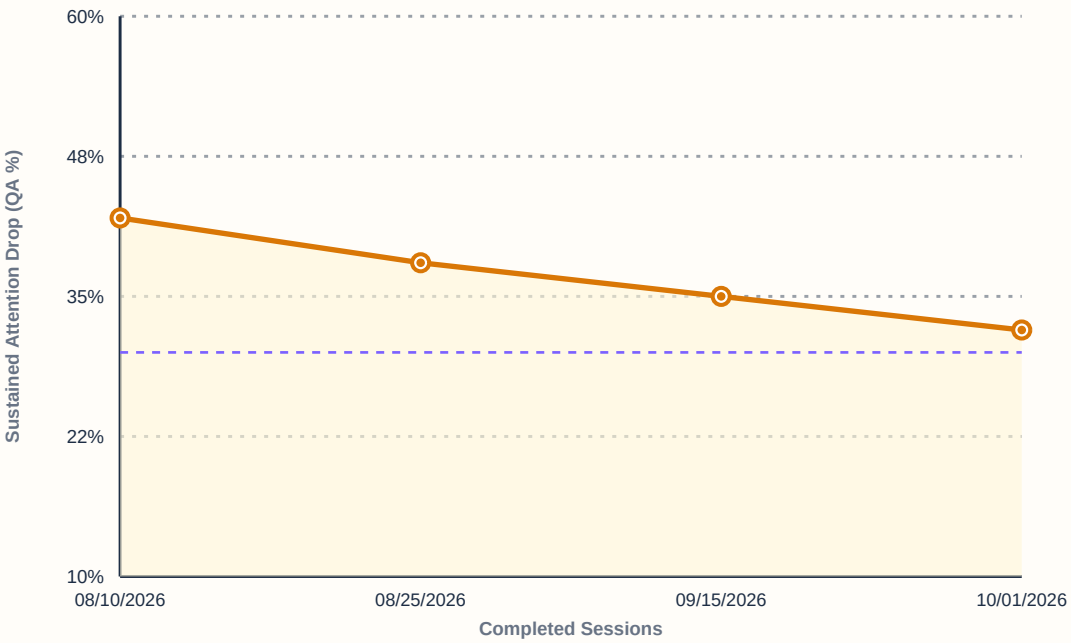
**Neurofunctional Analysis & Performance Opinion:**
Good overall engagement, with a tendency to respond hastily before inhibitory processing on No-Go stimuli. Across sessions, commission errors decreased from 6 to 4, demonstrating progressive motor deceleration adaptation under therapeutic guidance.

**Inhibitory Control Progression:** Longitudinal telemetry confirms progressive reduction in impulsive taps and automatic responses, indicating that motor inhibition mechanisms adapt satisfactorily when provided with structured slowing cues and response pacing.

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# Game Results

**Forest Photographer: Attentional Decline Index (QA) across Sessions**




**Clinical Synthesis & Evolution: Forest Photographer**

**Neurofunctional Analysis & Performance Opinion:**
Progressive accuracy decline from early minutes to session end, suggesting increased cognitive fatigue over time. Longitudinal tracking shows stabilization around QA 32%, confirming that structured breaks remain beneficial for sustained output.

**Attentional Endurance & Cognitive Fatigue:** The longitudinal trajectory highlights that sustained focus accuracy fluctuates as a function of continuous task duration, reinforcing the clinical necessity of strategic rest intervals to optimize attentional recharging in both academic and home environments.

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# Behavioral Synthesis

### 1. Response Variability

Response time oscillation highlights transient focus shifts typical of attentional fluctuation during repetitive tasks.

### 2. Inhibitory Control

Moderate motor deceleration difficulty observed, leading to hasty taps on stimuli requiring inhibition.

### 3. Sustained Attention

Downward curve in the second half of the session, compatible with early cognitive fatigue under sustained effort.

### 4. Joint Interpretation

The integrated profile shows that motor impulsivity associated with attentional fatigue is the child's main functional bottleneck.

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# Probabilistic Model

**Estimated Statistical Probability**

## 68% (High Probability of Attentive Profile)

**Assessment Date: 10/01/2026**

### Technical & Ethical Interpretation Notice

The reported statistical probability is derived from comparative predictive modeling based on digital telemetry biomarkers. **This result DOES NOT constitute a definitive medical or closed psychiatric diagnosis**. It serves as a standardized screening instrument that must be contextualized with qualitative clinical observations, in-person interviews, and developmental anamnesis conducted by a licensed healthcare professional.

### Biomarkers Rationale

The probabilistic model evaluates four foundational dimensions of the child's executive functioning: reaction latency variability (focus stability), impulsivity index (No-Go motor inhibition), sustained attention curve (fatigue across sequential blocks), and developmental chronological age.

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# Clinical Contextualization

### Detailed Anamnesis (development, school history, sleep, health, routine)

Motor and speech milestones achieved within expected chronological ranges. Mother reports restless sleep with occasional night awakenings. Persistent school complaints regarding task completion during continuous classroom instruction.

### Clinical Notes and Session Observations

Curious, highly communicative child with active sensory-seeking tendencies. Required verbal prompting and structure during stage transitions to maintain focus on the screen.

### Session History & Application Progress

Sequential sessions demonstrate positive motor adaptation curve while maintaining characteristic attentional oscillation during lengthy tasks.

# Integration of Findings

### Relationship between Telemetry, Anamnesis, and Clinical Notes

Marked convergence observed between telemetry performance decline in games and school reports regarding attentional fatigue and distractibility.

### Observed Convergences

Impulsivity detected during Tap Fast directly mirrors teacher reports of verbal impulsivity and turn-taking challenges in the classroom.

### Divergences & Preserved Domains

Preserved visuomotor coordination and playful problem-solving ability, indicating strong global cognitive potential without underlying motor deficits.

### Integrated Synthesis & Clinical Opinion

Overall clinical picture indicates a neuropsychological profile marked by inhibitory instability and attentional fatigability, in the absence of generalized intellectual impairment.

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# Professional Considerations

### Clinical Investigation & Therapeutic Focus

Investigate correlation between cognitive fatigue and demanding academic class periods.

### School & Pedagogical Adaptations

Implement pedagogical accommodations with long tasks segmented into 15-minute structured intervals.

### Monitoring & Re-evaluation Schedule

Neuropsychological monitoring and telemetry follow-up in 6 months to evaluate developmental progression.

### Complementary Standardized Scales

Administer complementary standardized rating scales completed by teachers and family (e.g., SNAP-IV).

### Routine, Sleep Hygiene & Lifestyle

Provide sleep hygiene counseling and evening routine structuring to support daytime attentional readiness.

# Conclusion

### Game Results

Performance characterized by moderate latency variability and No-Go commission error rate above chronological peer average.

### Probabilistic Result

Estimated probability of 68% for the evaluated attentional profile derived from the standardized screening model.

### Relationship with Clinical Context

High degree of consistency between collected digital biomarkers and qualitative behaviors observed in clinical consultation.

### Caveats & Professional Interpretation

This document represents a technical neuropsychological screening opinion and must be contextualized with comprehensive in-person qualitative assessment.

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# Signature & Traceability

**Responsible Psychologist**

Dr. Mariana Costa

**Professional License (CRP)**

CRP 06/142589

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**Authenticity & Technical Validity**

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